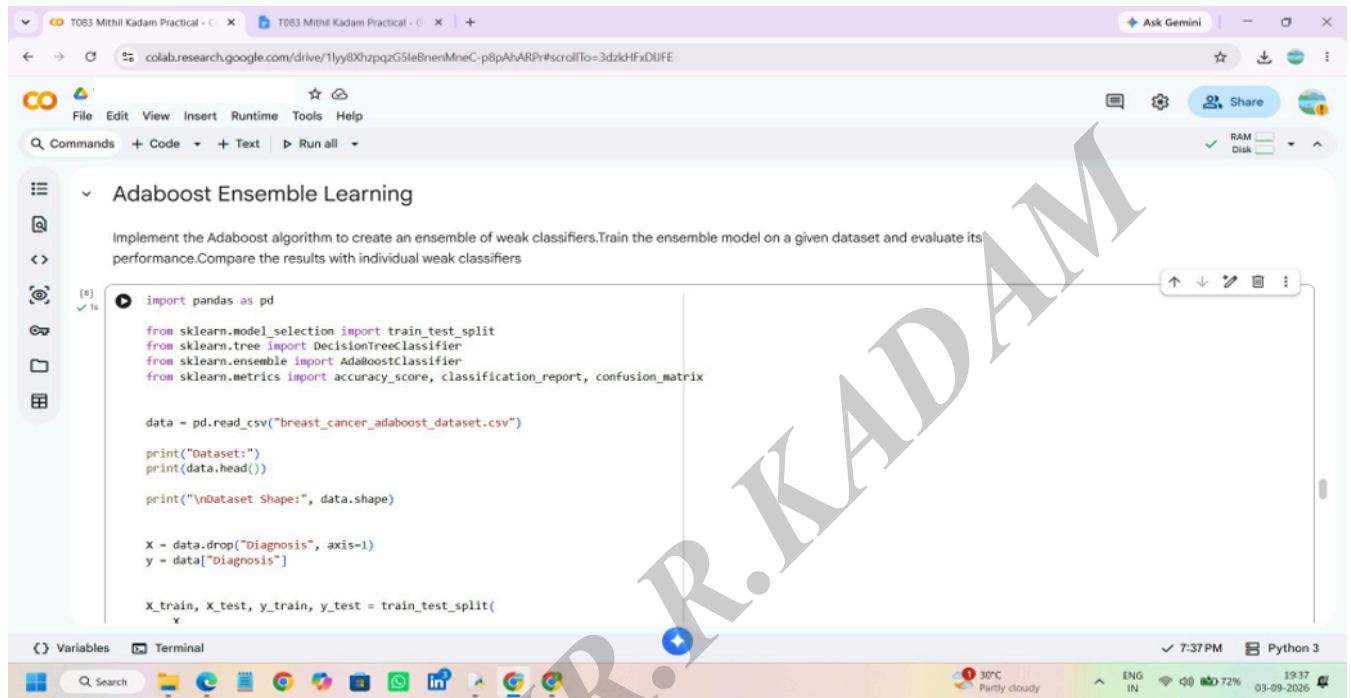


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Aim:- Adaboost Ensemble Learning

Implement the Adaboost algorithm to create an ensemble of weak classifiers. Train the ensemble model on a given dataset and evaluate its performance. Compare the results with individual weak classifiers



```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

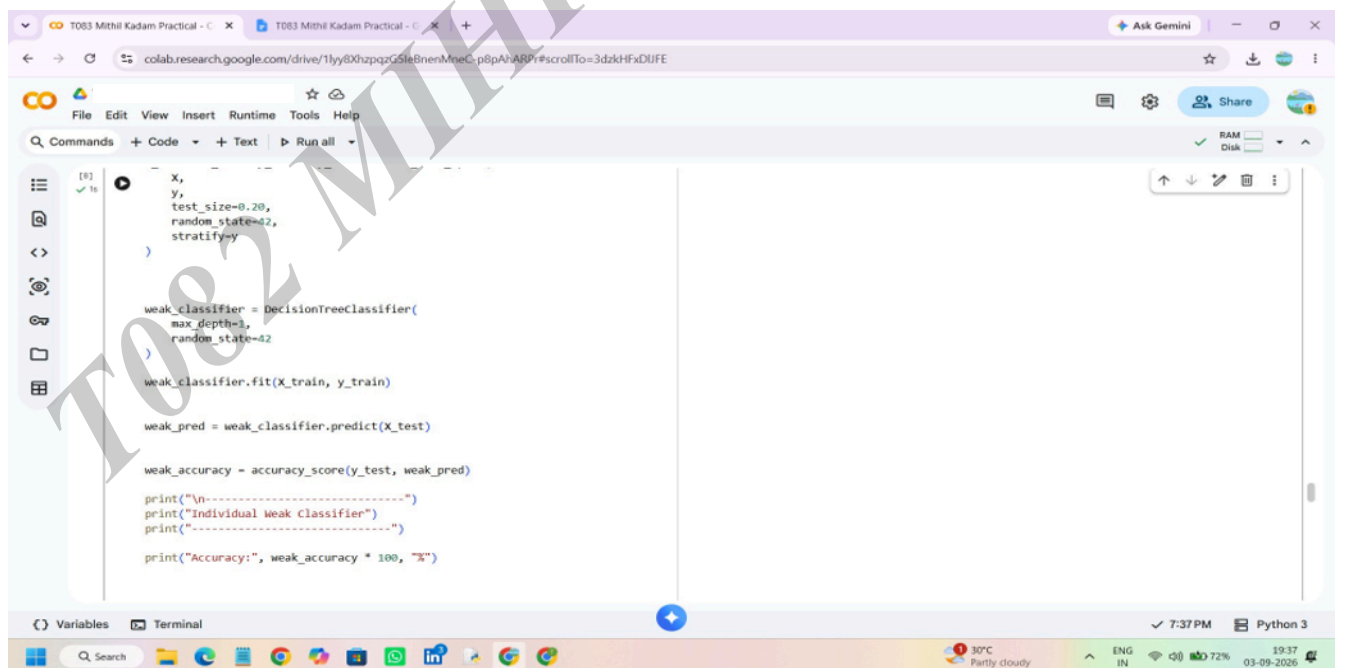
data = pd.read_csv("breast_cancer_adaboost_dataset.csv")

print("Dataset:")
print(data.head())

print("\nDataset Shape:", data.shape)

X = data.drop("Diagnosis", axis=1)
y = data["Diagnosis"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
```



```
X,
y,
test_size=0.20,
random_state=42,
stratify=y
)

weak_classifier = DecisionTreeClassifier(
    max_depth=3,
    random_state=42
)

weak_classifier.fit(X_train, y_train)

weak_pred = weak_classifier.predict(X_test)

weak_accuracy = accuracy_score(y_test, weak_pred)

print("\n-----")
print("Individual weak Classifier")
print("-----")

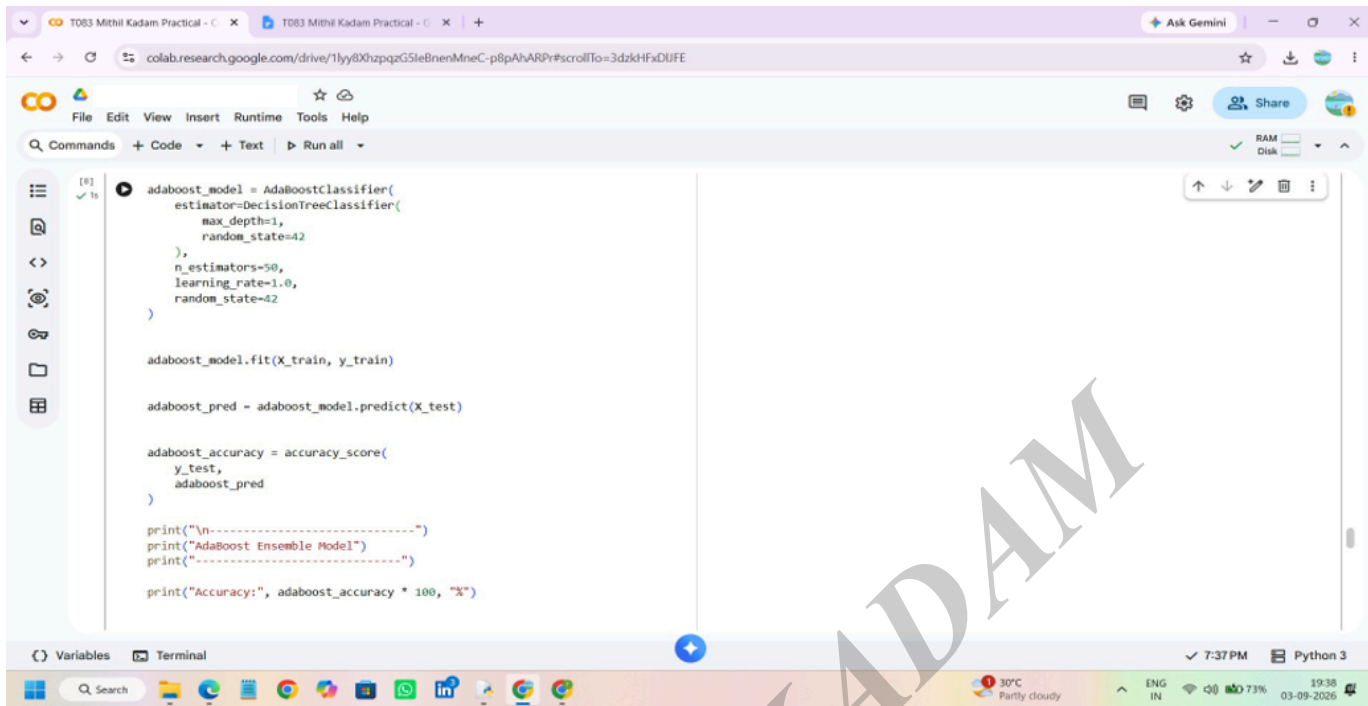
print("Accuracy:", weak_accuracy * 100, "%")
```

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The screenshot shows a Google Colab notebook with the following code:

```
[8] ✓ 1s
adaboost_model = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(
        max_depth=1,
        random_state=42
    ),
    n_estimators=50,
    learning_rate=1.0,
    random_state=42
)

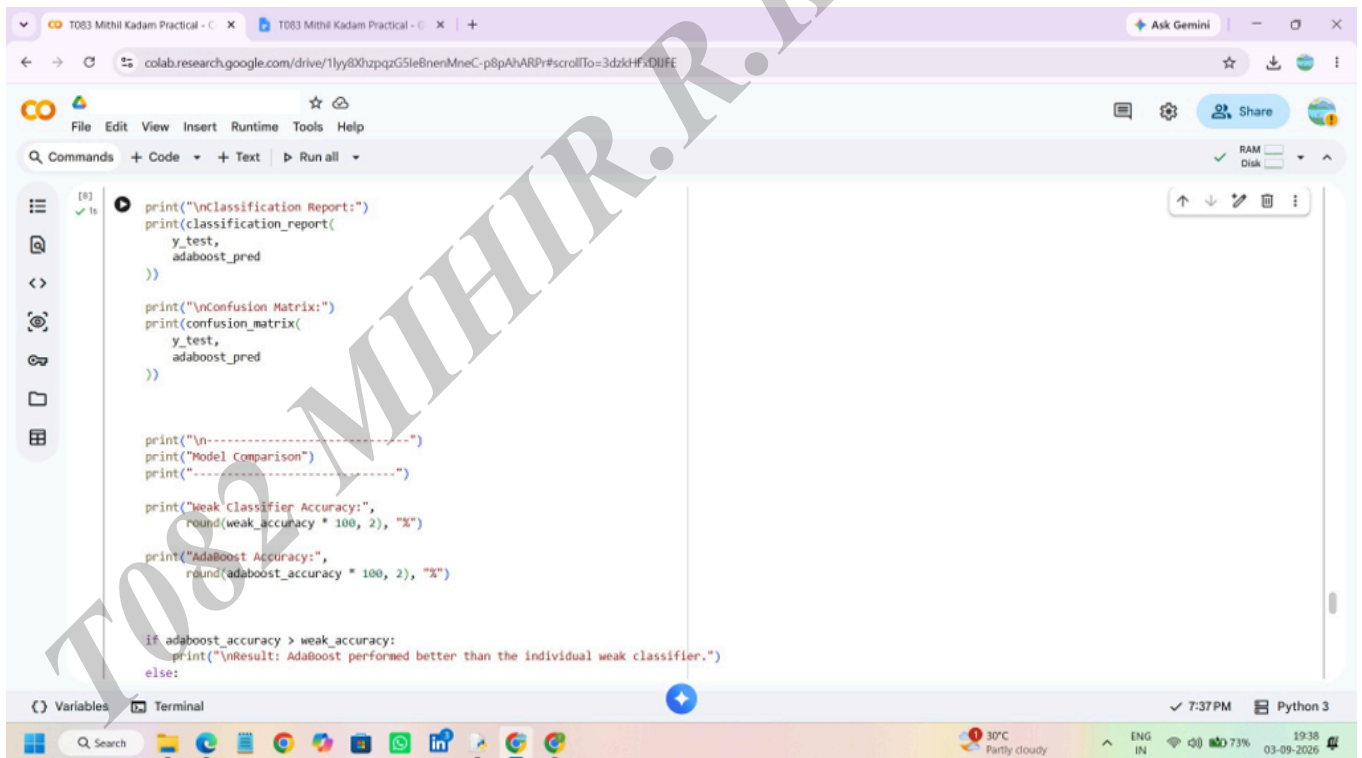
adaboost_model.fit(X_train, y_train)

adaboost_pred = adaboost_model.predict(X_test)

adaboost_accuracy = accuracy_score(
    y_test,
    adaboost_pred
)

print("\n-----")
print("AdaBoost Ensemble Model")
print("-----")
print("Accuracy:", adaboost_accuracy * 100, "%")
```

The interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for file operations, and a status bar at the bottom showing system information like temperature (30°C), battery (73%), and time (7:37 PM).



The screenshot shows the continuation of the Python script in the Google Colab notebook:

```
[9] ✓ 1s
print("\nClassification Report:")
print(classification_report(
    y_test,
    adaboost_pred
))

print("\nConfusion Matrix:")
print(confusion_matrix(
    y_test,
    adaboost_pred
))

print("\n-----")
print("Model Comparison")
print("-----")

print("Weak Classifier Accuracy:",
      round(weak_accuracy * 100, 2), "%")

print("AdaBoost Accuracy:",
      round(adaboost_accuracy * 100, 2), "%")

if adaboost_accuracy > weak_accuracy:
    print("\nResult: AdaBoost performed better than the individual weak classifier.")
else:
```

The interface is consistent with the previous screenshot, showing the same menu bar, toolbar, and status bar.

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```
if adaboost_accuracy > weak_accuracy:
    print("\nResult: AdaBoost performed better than the individual weak classifier.")
else:
    print("\nResult: AdaBoost did not perform better than the individual weak classifier.")

...
mean fractal dimension ... worst texture worst perimeter worst area \
0      0.07871 ...      17.33      184.60      2019.0
1      0.05667 ...      23.41      158.80      1956.0
2      0.05999 ...      25.53      152.50      1709.0
3      0.09744 ...      26.50      98.87      567.7
4      0.05883 ...      16.67      152.20      1575.0

worst smoothness worst compactness worst concavity worst concave points \
0      0.1622      0.6656      0.7119      0.2654
1      0.1238      0.1866      0.2416      0.1860
2      0.1444      0.4245      0.4504      0.2430
3      0.2098      0.8663      0.6869      0.2575
4      0.1374      0.2050      0.4000      0.1625

worst symmetry worst fractal dimension Diagnosis
0      0.4601      0.11890 Malignant
1      0.2750      0.08902 Malignant
2      0.3613      0.08758 Malignant
3      0.6638      0.17300 Malignant
4      0.2364      0.07678 Malignant

[5 rows x 31 columns]

Dataset Shape: (569, 31)
```

```
Dataset Shape: (569, 31)

...

Individual Weak Classifier
Accuracy: 87.71929824561403 %

AdaBoost Ensemble Model
Accuracy: 98.24561403508771 %

Classification Report:
precision    recall  f1-score   support

Benign      0.97      1.00      0.99        72
Malignant   1.00      0.95      0.98        42

accuracy          0.98        0.98        114
macro avg         0.99      0.98      0.98        114
weighted avg      0.98      0.98      0.98        114

Confusion Matrix:
[[72  0]
 [ 2 40]]

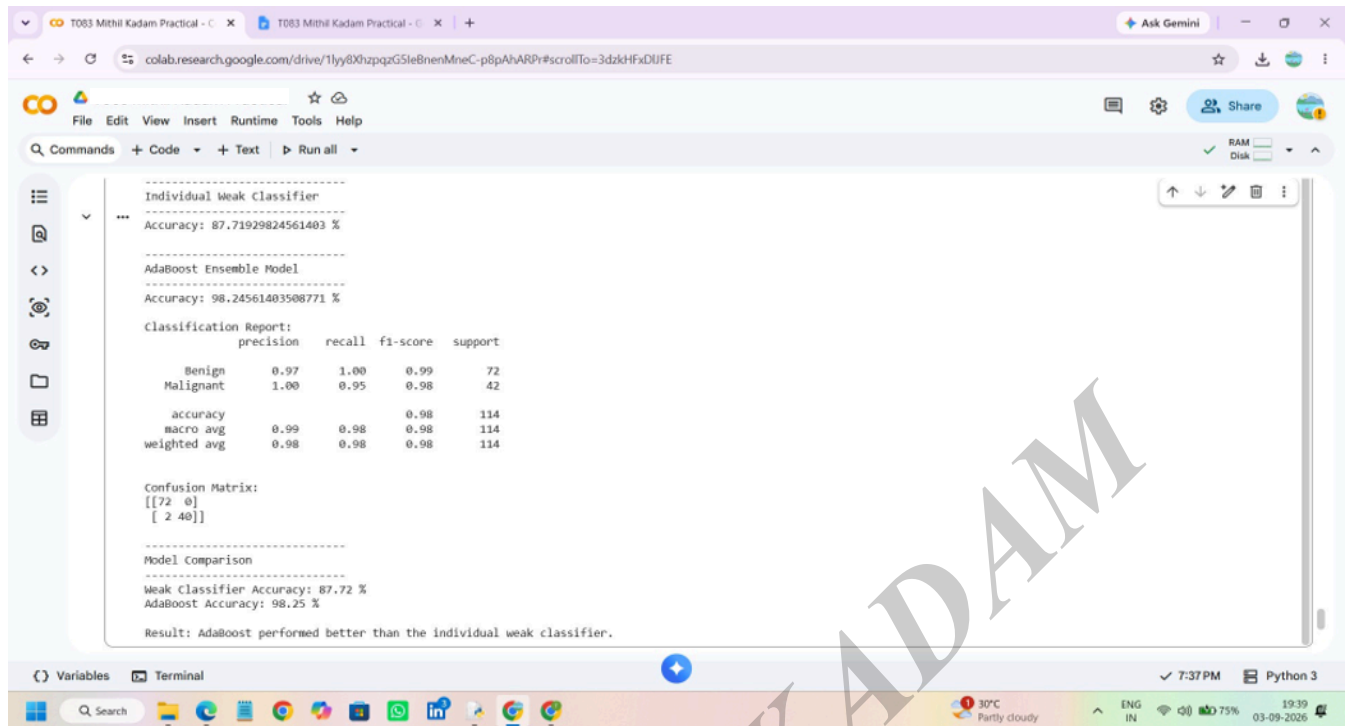
Model Comparison
Weak Classifier Accuracy: 87.72 %
AdaBoost Accuracy: 98.24 %
```

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The screenshot displays a Google Colab notebook interface. The main code cell contains the following output:

```
-----  
Individual Weak Classifier  
-----  
Accuracy: 87.71929824561403 %  
-----  
AdaBoost Ensemble Model  
-----  
Accuracy: 98.24561403508771 %  
-----  
Classification Report:  
precision    recall  f1-score   support  
  
   Benign    0.97    1.00    0.99        72  
  Malignant    1.00    0.95    0.98        42  
  
 accuracy    0.99    0.98    0.98       114  
  macro avg   0.99    0.98    0.98       114  
  weighted avg 0.98    0.98    0.98       114  
  
Confusion Matrix:  
[[72  0]  
 [ 2 40]]  
-----  
Model Comparison  
-----  
Weak Classifier Accuracy: 87.72 %  
AdaBoost Accuracy: 98.25 %  
  
Result: AdaBoost performed better than the individual weak classifier.
```

The interface includes a top navigation bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help' menus. A left sidebar contains icons for file management and execution. The bottom status bar shows 'Variables', 'Terminal', '7:37 PM', and 'Python 3'.

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